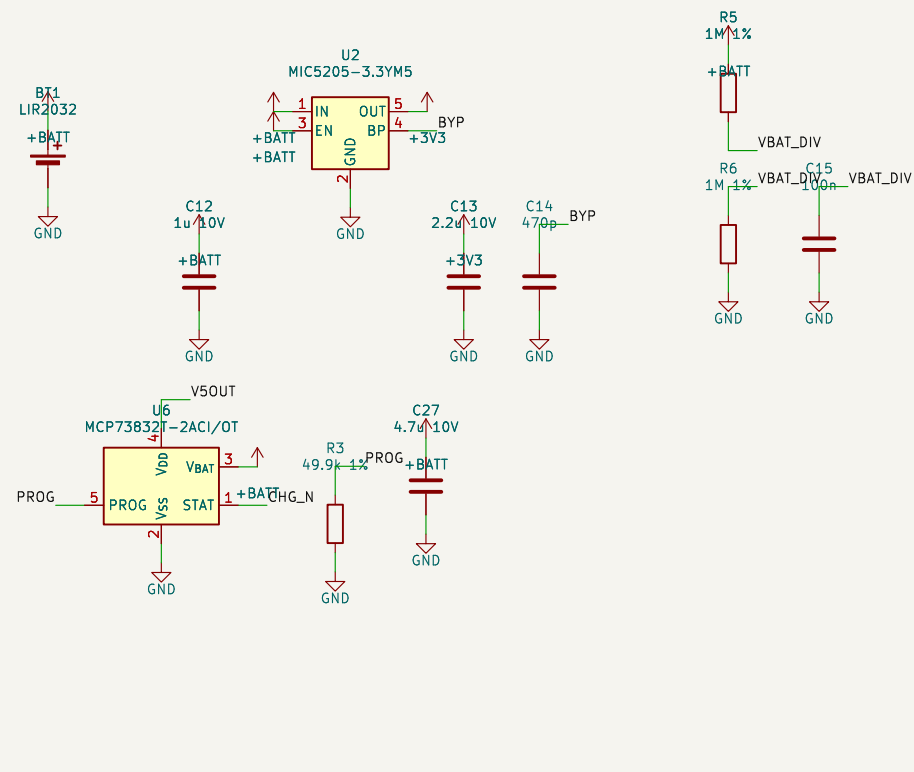
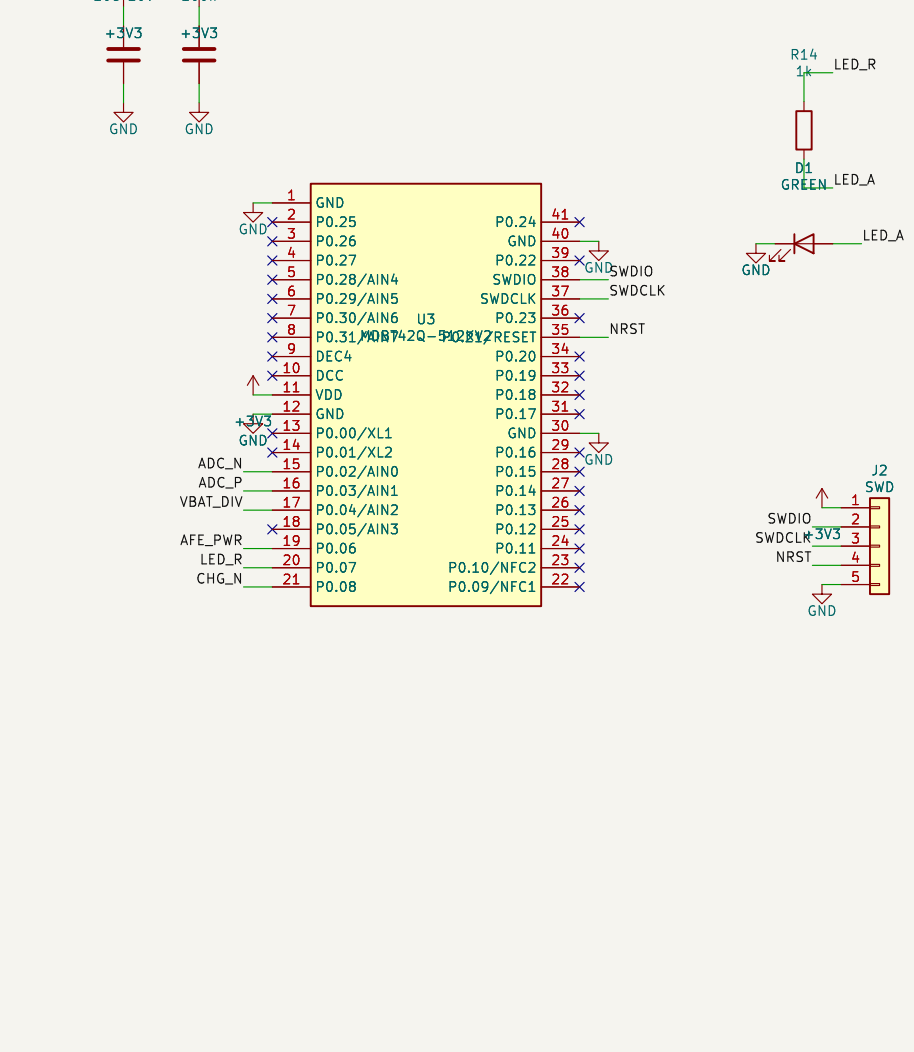




ANALOG FRONT-END: 2-OP-AMP POTENTIOSTAT (+0.512V)



DESIGN NOTES / VERIFY BEFORE LAYOUT

- 1. CHARGING: two-stage per TI guidance for small cells. BQ51013B = Qi receiver with regulated VS (VSOUT), ILIM: RLIM = 250/IMAX = R1+R2 = 5.25k -> IMAX 100mA (HW limit 120mA). MCP73832-2 (4.2V) charges the LIR2032: ICHG = 1000/RPROG = 1000/49.9k = 20mA (0.5C). termination -7.5% = 1.5mA, open-drain STAT -> CHG_N (100k pulled up to 3V3).
- This replaces the earlier BQ51050B direct-charge (TI: not recommended <20mA).
- 2. RESONANT TANK: Ls 11uH (TDK 202010-118W6-ID, 20x20x1.0mm, CRC 3500uH). C1 = (TAIPI * 100kHz)^2 = 230pF, 220n COG (fc 102kHz). C2 -> fd -1MHz -> 2.2n COG. Retune both whenever the coil changes. Small coil = tighter TX alignment; fine at our 0.13W.
- 3. MDBT420 pads 22-36 are a sequential reconstruction (P0.09/NFC1 ... P0.23); pads 1-21 and 37-41 verified against the datasheet. Every pad actually used in this design is a verified one. Cross-check the full pad table against the Raytac datasheet before PCB layout.
- 4. TS/CTRL: 10k NTC against the LIR2032 holder cuts wireless power if the cell heats (pin 10k = sense defeated, still a valid safe strap). EN1=EN2=GNDF; wireless enabled.
- 5. AFE_PWR is driven by nRF GPIO P0.06 set to high-drive. Load = REF35+OPA2391 - 60uA. Power the AFE off during runs at wireless electrode polarization.
- 6. RF: The quad rx driven from IREF35+OPA2391+OPA2391+OPA2391 TIA input, clean flux.
- 7. RF/CF options: 1M/470p (default, +2.1uA/-0.9uA FS), 5.1M/100p (427/181nA), 10M/47p (217/92nA). r_fm_ohm + oversample are runtime settings in firmware; recalibrate after a swap.
- 8. R10/C21 control-amp compensation are DNP; fit only if the CE-RE loop oscillates.
- 9. LIR2032 has no protection circuit: firmware must SYSTEM-OFF below 3.0V (VBAT_DIV, AIN2).
- 10. SAADC: differential P=AIN1(ADC_P) N=AIN0(ADC_N), gain 1/2.2, oversample 64x default.

1	2	3	4	5	6	7	8	9	10	11
OPA2391	REF35102	nRF52832(MDB42Q)	MIC5205	BQ51013B	MCP73832	LIR20	discrete-potentiostat	Sheet: /	File: discrete-potentiostat.kicad_sch	Title: Discrete Potentiostat - wireless 3-electrode chronoamperometry

Size: A2	Date: 2026-07-10	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 1/1	